

Impacto de los choques macroeconómicos sobre la actividad económica peruana: Un análisis SVAR con datos sintéticos (2015-2024)

[Autor 1]^{1*}

Autor 2

2

Autor 3

3

¹ Universidad Nacional del Altiplano, Facultad de Ingeniería Económica, Puno, Perú

² Docente del Curso de Econometría III, Universidad Nacional del Altiplano, Puno, Perú

³ Jefe de Práctica, Universidad Nacional del Altiplano, Facultad de Ingeniería Económica, Puno, Perú

*Autor de correspondencia: [email@example.com]

JEL Classification: C32; E62; E52; F41

Keywords: structural VAR; SVAR; macroeconomic shocks; impulse-response functions; variance decomposition; fiscal policy; monetary policy; Peru.

Abstract

Objective: This study analyzes the impact of macroeconomic shocks on Peruvian economic activity using a structural vector autoregressive (SVAR) model with synthetic monthly data from 2015 to 2024. **Theoretical Framework:** The research integrates Keynesian and monetarist perspectives to understand the dynamic transmission mechanisms of fiscal and monetary policy shocks. **Method:** Six macroeconomic variables were employed: Gross Domestic Product (GDP), inflation rate, nominal exchange rate, reference interest rate, money supply (M2), and government spending. Structural shock identification was performed through Cholesky decomposition with a recursive ordering grounded in economic theory. The econometric procedure included unit root tests (ADF, PP, KPSS), optimal lag selection (AIC, BIC, HQIC, FPE), VAR stability verification, impulse-response functions (IRF), forecast error variance decomposition (FEVD), and historical decomposition. **Results:** The impulse-response functions reveal that government spending shocks generate a positive and persistent effect on GDP, with a maximum impact of 0.72 % observed in the third month. Interest rate shocks show a contractionary effect of up to -0.38 %. The FEVD indicates that government spending and money supply jointly explain approximately 45 % of GDP variability in 12-month horizons. Historical decomposition shows that economic policy shocks significantly contributed to GDP fluctuations during the post-pandemic recovery period. **Originality/Value:** Although based on simulated data for academic purposes, this study demonstrates the applicability of the SVAR approach for macroeconomic dynamics analysis and provides a robust methodological framework applicable to real data in emerging economies.

Keywords: structural VAR; SVAR; macroeconomic shocks; impulse-response functions; variance decomposition; fiscal policy; monetary policy; Peru.

JEL Classification: C32; E62; E52; F41.

Resumen

Objetivo: El presente estudio analiza el impacto de los choques macroeconómicos sobre la actividad económica peruana mediante un modelo SVAR con datos sintéticos mensuales para el período 2015-2024. **Marco Teórico:** La investigación integra las perspectivas keynesiana y monetarista para comprender los mecanismos de transmisión dinámica de los choques de política fiscal y monetaria. **Método:** Se emplearon seis variables macroeconómicas: PIB, tasa de inflación, tipo de cambio nominal, tasa de interés de referencia, oferta monetaria (M2) y gasto público. La identificación de los choques estructurales se realizó mediante descomposición de Cholesky. El procedimiento econométrico incluyó pruebas de raíz unitaria, selección óptima de rezagos, verificación de estabilidad del VAR, funciones impulso-respuesta (IRF), descomposición de varianza (FEVD) y descomposición histórica. **Resultados:** Las IRF revelan que los

51 choques de gasto público generan un efecto positivo y persistente sobre el PIB (máximo 0.72 %
52 en $t+3$), mientras que los choques de tasa de interés muestran un efecto contractivo (-0.38 %).
53 La FEVD indica que el gasto público y la oferta monetaria explican conjuntamente el 45 % de
54 la variabilidad del PIB a 12 meses. **Originalidad/Valor:** El estudio demuestra la aplicabilidad
55 del enfoque SVAR para el análisis de dinámica macroeconómica en economías emergentes.

56 **Palabras clave:** VAR estructural; SVAR; choques macroeconómicos; funciones impulso-respuesta;
57 descomposición de varianza; política fiscal; política monetaria; Perú.

1. Introduction

The analysis of macroeconomic shock transmission mechanisms represents one of the fundamental challenges of modern macroeconomics. Since the seminal contribution of Sims (1980), vector autoregressive (VAR) models have become an indispensable tool for studying the dynamic relationships among macroeconomic variables. The structural VAR (SVAR) approach, which incorporates theoretical restrictions to identify structural shocks, has enabled researchers to analyze the causal effects of economic policy interventions and external disturbances on the macroeconomy (Bernanke y Blinder, 1992; Blanchard y Perotti, 2002; Christiano, Eichenbaum, y Evans, 1999; Kilian y Lütkepohl, 2017; Uhlig, 2005).

In the Latin American context, a growing body of literature has applied SVAR methodology to examine the effects of monetary and fiscal policy on economic activity. C'espedes y Chang (2020) analyzed monetary policy transmission in dollarized economies, finding that the credit channel operates with significant limitations in contexts of high financial dollarization. Castillo, Montoro, y Tuesta (2020) examined the effects of commodity price shocks on regional economies, demonstrating that external volatility constitutes a relevant source of output fluctuations. Fern'andez, Gonz'alez, y Rodr'iguez (2021) evaluated fiscal policy effectiveness in middle-income countries, concluding that fiscal multipliers are significantly smaller in economies with high labor informality. Moreno-Brid y S'anchez (2019) documented the importance of external shocks in Latin American growth dynamics, while Ocampo y Parra (2020) analyzed the relationship between monetary policy and financial stability in the region.

For the Peruvian case, empirical research on macroeconomic shocks has gained relevance in the last two decades, particularly following the implementation of the inflation targeting framework and the adoption of fiscal rules. Mendoza y Colich'on (2018) analyzed the effects of monetary policy on output and inflation in Peru, finding that interest rate shocks have significant although transitory effects on economic activity. Castillo y Winkelried (2019) evaluated the transmission of external shocks to the Peruvian economy, demonstrating that terms of trade constitute the most relevant channel for international fluctuation transmission. Humala y Rodr'iguez (2020) examined the relationship between fiscal policy and the business cycle in Peru, identifying a procyclical behavior of government spending during expansion periods. Dancourt (2021) analyzed the limits of monetary policy in contexts of partial financial dollarization, while Le'on y Quispe (2022) studied the transmission of exchange rate shocks to domestic inflation.

The research problem addressed by this study relates to the need to understand the transmission mechanisms of macroeconomic shocks in the Peruvian context. Despite advances in the empirical literature, questions remain regarding the magnitude and persistence of fiscal and monetary policy effects on economic activity, as well as the relative importance of different types of structural shocks in generating output fluctuations. This issue is particularly relevant in a post-pandemic context, where economic policymakers face the challenge of designing ma-

106 macroeconomic stabilization strategies in an environment of elevated uncertainty.

107 The research gap identified lies in the scarcity of studies that comprehensively analyze
108 the interaction between fiscal, monetary, and real shocks in the Peruvian economy using an
109 SVAR framework with a broad set of variables. While previous works have examined specific
110 transmission channels, limited studies provide a systemic view of Peruvian macroeconomic
111 dynamics incorporating fiscal policy, monetary policy, and real sector variables simultaneously.
112 In this sense, the present study contributes to filling this gap by applying an SVAR model that
113 allows identifying and quantifying the effects of different structural shocks on the economic
114 system.

115 The theoretical justification of the study is grounded in the need to integrate Keynesian and
116 monetarist approaches to understand macroeconomic dynamics. From the Keynesian perspec-
117 tive, aggregate demand shocks, particularly government spending, have significant effects on
118 output in the short run, especially in contexts of slack capacity and nominal rigidities (Auer-
119 bach y Gorodnichenko, 2012; Blanchard y Perotti, 2002; Fat'as y Mihov, 2001; Parker, 2011;
120 C. Romer y Bernstein, 2009). From the monetarist perspective, money supply and interest rate
121 shocks constitute the main determinants of output and inflation fluctuations in the medium run
122 (Barro, 1977; Friedman, 1968; Lucas, 1972; McCallum, 1989; Sargent y Wallace, 1975).

123 The general objective of this study is to analyze the impact of macroeconomic shocks on
124 Peruvian economic activity during the 2015-2024 period through the estimation of a structur-
125 al VAR model that allows identifying transmission mechanisms and quantifying the relative
126 importance of each type of shock in GDP fluctuations. Specifically, we seek to: (i) estimate
127 impulse-response functions to evaluate the magnitude, direction, and persistence of structural
128 shock effects; (ii) perform forecast error variance decomposition to determine the relative con-
129 tribution of each shock to the variability of endogenous variables; and (iii) conduct historical
130 decomposition to analyze the contribution of structural shocks to the temporal evolution of
131 macroeconomic variables. The central hypothesis holds that government spending and money
supply shocks constitute the main sources of GDP fluctuation in the short run, while interest
rate and exchange rate shocks have significant although transitory effects.

124 **2. Theoretical Framework**

125 **2.1. Keynesian Theory**

126 Keynesian theory constitutes the central theoretical foundation of this study. According to
127 Keynes (1936), economic fluctuations are fundamentally the result of variations in aggregate
128 demand, which can originate from changes in consumption, investment, government spending,
129 or net exports. The fiscal multiplier plays a central role by amplifying the initial impact of
130 a change in government spending on output, generating indirect effects through consumption
131 induced by increased disposable income (Blanchard, 2017; Eggertsson y Krugman, 2012; Man-

kiw, 2020; D. Romer, 2019; Woodford, 2011). Keynesian mechanisms have been widely validated in contexts of liquidity traps and nominal rigidities, where fiscal policy acquires special effectiveness.

2.2. Monetarist Theory

From the monetarist perspective, developed by Friedman (1968) and subsequently refined by Lucas (1972), the money supply constitutes the main determinant of output fluctuations in the short run and inflation in the long run. The monetary transmission mechanism operates through multiple channels, including the interest rate channel, the credit channel, the exchange rate channel, and the expectations channel (Bernanke y Gertler, 1995; Clarida, Gal'i, y Gertler, 1999; Mishkin, 1995; Svensson, 2010; Taylor, 1995).

2.3. Theory of Structural Shocks

Structural shocks represent exogenous unanticipated disturbances that affect the economic system and have permanent or transitory effects on macroeconomic variables. Within the SVAR framework, structural shocks are classified according to their origin and nature, distinguishing between supply, demand, monetary, fiscal, technological, and external shocks (Blanchard y Quah, 1989; Cover, 1992; Gal'i, 1999; Mountford y Uhlig, 2009; Shapiro y Watson, 1988).

2.4. Transmission Mechanisms

Transmission mechanisms describe the channels through which structural shocks propagate to the rest of the economic system. For fiscal shocks, the main mechanism operates through the government spending multiplier (Auerbach y Gorodnichenko, 2012; Blanchard y Perotti, 2002; Fat'as y Mihov, 2001; Hall, 2009; Ramey, 2011). For monetary shocks, transmission channels include the interest rate channel, the credit channel, and the exchange rate channel.

2.5. SVAR Model Framework

The structural VAR model extends the conventional VAR approach by incorporating theoretical restrictions that allow recovering the underlying structural shocks from the reduced-form innovations (Bernanke, 1986; Blanchard y Watson, 1986; Christiano y cols., 1999; Kilian y Lütkepohl, 2017; Sims, 1980). The general reduced-form VAR(p) is specified as:

$$Y_t = c + \Phi_1 Y_{t-1} + \dots + \Phi_p Y_{t-p} + u_t \quad (1)$$

where $Y_t = [G_t, M2_t, GDP_t, INFL_t, ER_t, IR_t]'$ is the vector of endogenous variables, Φ_i are 6×6 coefficient matrices, and $u_t \sim N(0, \Sigma_u)$.

The structural form is specified as:

$$AY_t = c^* + \Phi_1^* Y_{t-1} + \dots + \Phi_p^* Y_{t-p} + B\varepsilon_t \quad (2)$$

where A is a matrix of contemporaneous impact coefficients, B is a diagonal matrix, and ε_t are structural shocks with $E[\varepsilon_t \varepsilon_t'] = I$.

3. Methodology

3.1. Research Design

This study corresponds to explanatory research with a non-experimental, longitudinal, and retrospective design. A quantitative approach based on econometric time series analysis is employed, using monthly macroeconomic data for the 2015-2024 period.

3.2. Variables and Data

Six macroeconomic variables are considered: Gross Domestic Product (GDP), inflation rate (INFL), nominal exchange rate (ER), reference interest rate (IR), money supply (M2) in logarithms, and government spending (G). The data are synthetic in nature, generated through stochastic processes calibrated to replicate the statistical properties of Peruvian macroeconomic series. The database contains 120 monthly observations from January 2015 to December 2024.

3.3. Econometric Procedure

The econometric procedure follows several stages: (i) stationarity tests (ADF, PP, KPSS); (ii) optimal lag selection (AIC, BIC, HQIC, FPE); (iii) reduced-form VAR estimation; (iv) VAR stability verification; (v) structural shock identification via Cholesky decomposition; (vi) impulse-response functions; (vii) forecast error variance decomposition; and (viii) historical decomposition.

3.4. Identification Strategy

The identification is performed through Cholesky decomposition imposing a recursive structure with the ordering: $G \rightarrow M2 \rightarrow GDP \rightarrow INFL \rightarrow ER \rightarrow IR$. Government spending does not respond contemporaneously to shocks from other variables. The money supply responds to fiscal shocks but not to real or financial shocks. GDP responds to fiscal and monetary shocks. Inflation responds to fiscal, monetary, and output shocks. The exchange rate responds to all variables except the interest rate. The interest rate responds contemporaneously to all shocks.

4. Results

4.1. Descriptive Statistics

Descriptive analysis reveals that GDP presented an average value of 102.47 points with a standard deviation of 2.85 points. The average inflation rate stood at 3.52 % (SD = 0.98 %). The nominal exchange rate showed an average of 3.82 S/./USD. The reference interest rate presented a mean value of 5.12 %.

4.2. Stationarity Tests

All series present a unit root in levels (I(1)). First differences were stationary at the 5 % significance level, consistent with Nelson y Plosser (1982) and Perron (1989).

4.3. Optimal Lag Selection

The AIC suggested a VAR(4), while BIC and HQIC selected a VAR(2). Following Lütkepohl (2005) and Kilian y Lütkepohl (2017), a VAR(4) was chosen.

4.4. VAR Stability

All inverse roots lie within the unit circle (maximum modulus: 0.89), confirming model stability.

4.5. Impulse-Response Functions

A positive one-standard-deviation shock to government spending generates an immediate GDP increase of 0.45 %, reaching a maximum of 0.72 % in the third month, converging to equilibrium in approximately 18 months. An unanticipated increase in the reference interest rate generates a contractionary effect on GDP, with a maximum decline of 0.38 % between the fourth and sixth month.

Figura 1: Impulse-Response Functions



Response of GDP and Inflation to selected structural shocks.

4.6. Forecast Error Variance Decomposition

For GDP at a 12-month horizon, government spending explains 28.5 % of output variability, followed by money supply (16.3 %), inflation (12.1 %), interest rate (9.8 %), exchange rate (7.2 %), and own GDP innovation (26.1 %).

4.7. Historical Decomposition

Government spending shocks contributed positively to GDP growth during the 2021-2022 post-pandemic recovery, while contractionary monetary policy shocks during 2023 had a nega-

tive effect on output.

5. Discussion

The results provide evidence on the importance of fiscal and monetary shocks in Peruvian macroeconomic dynamics. The positive response of GDP to government spending shocks is consistent with Blanchard y Perotti (2002) and Auerbach y Gorodnichenko (2012). Christiano y cols. (1999) and Bernanke y Blinder (1992) documented similar contractionary effects of interest rate shocks in the United States.

The variance decomposition results are consistent with Stock y Watson (2001), who found that economic policy shocks explain an important fraction of output fluctuations. Primiceri (2005) documented changes in the relative importance of shocks over time, while Nakajima (2011) developed Bayesian methods for time-varying parameter VARs.

The economic policy implications suggest that authorities can employ both fiscal and monetary instruments for stabilization. C. Romer y Bernstein (2009) and Parker (2011) demonstrated the effectiveness of well-designed fiscal stimuli during recessions. Woodford (2011) and Blanchard (2017) provided theoretical frameworks for optimal fiscal and monetary policy coordination.

6. Conclusion

This study analyzed the impact of macroeconomic shocks on Peruvian economic activity using an SVAR model. The results confirm that government spending and money supply shocks constitute the main sources of output fluctuation. The main findings are: (i) fiscal shock effects are more persistent than monetary shock effects; (ii) government spending and money supply jointly explain approximately 45 % of GDP variability; (iii) fiscal policy shocks had a significant contribution during the post-pandemic recovery; and (iv) the SVAR model satisfies stability conditions, ensuring valid inference.

Future research could extend this analysis by incorporating time-varying parameters, using sign restrictions for identification, or applying the methodology to real data from the Peruvian economy.

Referencias

- Auerbach, A. J., y Gorodnichenko, Y. (2012). Measuring the output responses to fiscal policy. *American Economic Journal: Economic Policy*, 4(2), 1-27.
- Barro, R. J. (1977). Unanticipated money growth and unemployment in the united states. *American Economic Review*, 67(2), 101-115.
- Bernanke, B. S. (1986). Alternative explanations of the money-income correlation. En *Carnegie-rochester conference series on public policy* (Vol. 25, p. 49-99).
- Bernanke, B. S., y Blinder, A. S. (1992). The federal funds rate and the channels of monetary transmission. *American Economic Review*, 82(4), 901-921.
- Bernanke, B. S., y Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4), 27-48.
- Blanchard, O. (2017). The need for different fiscal policies. *Journal of Economic Perspectives*, 31(1), 3-20.
- Blanchard, O., y Perotti, R. (2002). An empirical characterization of the dynamic effects of changes in government spending and taxes on output. *Quarterly Journal of Economics*, 117(4), 1329-1368.
- Blanchard, O., y Quah, D. (1989). The dynamic effects of aggregate demand and supply disturbances. *American Economic Review*, 79(4), 655-673.
- Blanchard, O., y Watson, M. W. (1986). Are business cycles all alike? En R. J. Gordon (Ed.), *The american business cycle* (p. 123-180). University of Chicago Press.
- Castillo, P., Montoro, C., y Tuesta, V. (2020). Commodity prices and macroeconomic dynamics in emerging economies. *Journal of International Money and Finance*, 101, 102-125.
- Castillo, P., y Winkelried, D. (2019). External shocks and macroeconomic dynamics in peru. *Revista Estudios Económicos*, 37, 9-30.
- C'espedes, L. F., y Chang, R. (2020). Monetary policy in dollarized economies. *Journal of International Economics*, 125, 103-128.
- Christiano, L. J., Eichenbaum, M., y Evans, C. L. (1999). Monetary policy shocks: What have we learned and to what end? En J. B. Taylor y M. Woodford (Eds.), *Handbook of macroeconomics* (Vol. 1, p. 65-148). Elsevier.
- Clarida, R., Gal'i, J., y Gertler, M. (1999). The science of monetary policy: A new keynesian perspective. *Journal of Economic Literature*, 37(4), 1661-1707.
- Cover, J. P. (1992). Asymmetric effects of positive and negative money-supply shocks. *Quarterly Journal of Economics*, 107(4), 1261-1282.
- Dancourt, O. (2021). L'imites de la pol'itica monetaria en econom'ias dolarizadas. *Revista Econom'ia*, 44(87), 5-28.
- Eggertsson, G. B., y Krugman, P. (2012). Debt, deleveraging, and the liquidity trap: A fisher-minsky-koo approach. *Quarterly Journal of Economics*, 127(3), 1469-1513.
- Fat'as, A., y Mihov, I. (2001). Government size and automatic stabilizers: International evi-

- dence. *Journal of International Economics*, 55(1), 3-28.
- Fernández, A., González, A., y Rodríguez, D. (2021). Fiscal multipliers in middle-income countries. *Journal of Development Economics*, 150, 102-128.
- Friedman, M. (1968). The role of monetary policy. *American Economic Review*, 58(1), 1-17.
- Galí, J. (1999). Technology, employment, and the business cycle: Do technology shocks explain aggregate fluctuations? *American Economic Review*, 89(1), 249-271.
- Hall, R. E. (2009). By how much does gdp rise if the government buys more output? *Brookings Papers on Economic Activity*, 40(2), 183-249.
- Humala, A., y Rodríguez, G. (2020). Política fiscal y ciclo económico en Perú. *Revista Estudios Económicos*, 39, 31-52.
- Keynes, J. M. (1936). *The general theory of employment, interest and money*. Macmillan.
- Kilian, L., y Lütkepohl, H. (2017). *Structural vector autoregressive analysis*. Cambridge University Press.
- León, J., y Quispe, Z. (2022). Transmisión de choques cambiarios a la inflación en Perú. *Revista Moneda*, 189, 14-22.
- Lucas, R. E. (1972). Expectations and the neutrality of money. *Journal of Economic Theory*, 4(2), 103-124.
- Lütkepohl, H. (2005). *New introduction to multiple time series analysis*. Springer.
- Mankiw, N. G. (2020). *Macroeconomics* (10.^a ed.). Worth Publishers.
- McCallum, B. T. (1989). *Monetary economics: Theory and policy*. Macmillan.
- Mendoza, O., y Colichón, L. (2018). Efectos de la política monetaria sobre el producto y la inflación en Perú. *Revista Estudios Económicos*, 35, 45-68.
- Mishkin, F. S. (1995). Symposium on the monetary transmission mechanism. *Journal of Economic Perspectives*, 9(4), 3-10.
- Moreno-Brid, J. C., y Sánchez, J. (2019). External shocks and growth dynamics in latin america. *CEPAL Review*, 128, 7-28.
- Mountford, A., y Uhlig, H. (2009). What are the effects of fiscal policy shocks? *Journal of Applied Econometrics*, 24(6), 960-992.
- Nakajima, J. (2011). Time-varying parameter VAR model with stochastic volatility. *Monetary and Economic Studies*, 29, 107-142.
- Nelson, C. R., y Plosser, C. I. (1982). Trends and random walks in macroeconomic time series. *Journal of Monetary Economics*, 10(2), 139-162.
- Ocampo, J. A., y Parra, M. (2020). Monetary policy and financial stability in latin america. *Journal of Post Keynesian Economics*, 43(1), 1-28.
- Parker, J. A. (2011). On measuring the effects of fiscal policy in recessions. *Journal of Economic Literature*, 49(3), 703-718.
- Perron, P. (1989). The great crash, the oil price shock, and the unit root hypothesis. *Econometrica*, 57(6), 1361-1401.
- Primiceri, G. E. (2005). Time varying structural vector autoregressions and monetary policy.

321 *Review of Economic Studies*, 72(3), 821-852.

322 Ramey, V. A. (2011). Identifying government spending shocks: It is all in the timing. *Quarterly*
323 *Journal of Economics*, 126(1), 1-50.

324 Romer, C., y Bernstein, J. (2009). *The job impact of the american recovery and reinvestment*
325 *plan* (Inf. Téc.). Council of Economic Advisers.

326 Romer, D. (2019). *Advanced macroeconomics* (5.^a ed.). McGraw-Hill.

327 Sargent, T. J., y Wallace, N. (1975). Rational expectations, the optimal monetary instrument,
328 and the optimal money supply rule. *Journal of Political Economy*, 83(2), 241-254.

329 Shapiro, M. D., y Watson, M. W. (1988). Sources of business cycle fluctuations. *NBER*
330 *Macroeconomics Annual*, 3, 111-148.

331 Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1), 1-48.

332 Stock, J. H., y Watson, M. W. (2001). Vector autoregressions. *Journal of Economic Perspecti-*
333 *ves*, 15(4), 101-115.

334 Svensson, L. E. O. (2010). Inflation targeting. En B. M. Friedman y M. Woodford (Eds.),
335 *Handbook of monetary economics* (Vol. 3, p. 1237-1302). Elsevier.

336 Taylor, J. B. (1995). The monetary transmission mechanism: An empirical framework. *Journal*
337 *of Economic Perspectives*, 9(4), 11-26.

338 Uhlig, H. (2005). What are the effects of monetary policy on output? results from an agnostic
339 identification procedure. *Journal of Monetary Economics*, 52(2), 381-419.

340 Woodford, M. (2011). Simple analytics of the government spending multiplier. *American*
341 *Economic Journal: Macroeconomics*, 3(1), 1-35.

A. Structural Identification Matrices

This appendix presents the matrices used for structural identification via Cholesky decomposition.

Cuadro 1: Contemporaneous Impact Matrix (Cholesky Decomposition)

Shock \ Var	G	M2	GDP	INFL	ER	IR
ε_G	0.1842	0.0000	0.0000	0.0000	0.0000	0.0000
ε_{M2}	0.0421	0.1537	0.0000	0.0000	0.0000	0.0000
ε_{GDP}	0.0385	0.0294	0.1278	0.0000	0.0000	0.0000
ε_{INFL}	0.0213	0.0182	0.0351	0.1124	0.0000	0.0000
ε_{ER}	0.0158	0.0127	0.0283	0.0196	0.0983	0.0000
ε_{IR}	0.0326	0.0245	0.0412	0.0368	0.0281	0.0895

Recursive Cholesky ordering: $G \rightarrow M2 \rightarrow GDP \rightarrow INFL \rightarrow ER \rightarrow IR$.

Cuadro 2: Structural Shocks Standard Deviations (B Matrix)

Shock	Std. Deviation	Variance
ε_G	0.1842	0.03394
ε_{M2}	0.1596	0.02547
ε_{GDP}	0.1365	0.01863
ε_{INFL}	0.1221	0.01491
ε_{ER}	0.1058	0.01120
ε_{IR}	0.1123	0.01261

B. Supplementary Tables

Cuadro 3: Augmented Dickey-Fuller Test Results

Variable	ADF Stat.	p-value	1 % Crit.	5 % Crit.	Decision
PIB	−1,2842	0.6321	−3,4865	−2,8861	No estacionaria
INFLACION	−1,9563	0.3047	−3,4865	−2,8861	No estacionaria
TIPO_CAMBIO	−1,0187	0.7483	−3,4865	−2,8861	No estacionaria
TASA_INTERES	−1,8732	0.3421	−3,4865	−2,8861	No estacionaria
OFERTA_MONETARIA	−0,8476	0.8124	−3,4865	−2,8861	No estacionaria
GASTO_GOBIERNO	−1,4521	0.5587	−3,4865	−2,8861	No estacionaria
D(PIB)	−5,8342	0.0000	−3,4865	−2,8861	Estacionaria
D(INFLACION)	−6,1247	0.0000	−3,4865	−2,8861	Estacionaria
D(TIPO_CAMBIO)	−5,4238	0.0001	−3,4865	−2,8861	Estacionaria
D(TASA_INTERES)	−6,8913	0.0000	−3,4865	−2,8861	Estacionaria
D(OFERTA_MONETARIA)	−5,2189	0.0001	−3,4865	−2,8861	Estacionaria
D(GASTO_GOBIERNO)	−6,4528	0.0000	−3,4865	−2,8861	Estacionaria

D(·) denotes first difference. All tests include constant term.

Cuadro 4: Optimal Lag Selection Criteria

Lag	AIC	BIC	HQIC	FPE
0	4.2831	4.3524	4.3118	72.5842
1	2.1847	2.4629	2.2984	8.9147
2	1.8245	2.3116	2.0232	6.2145
3	1.6932	2.3892	1.9769	5.4382
4	1.5218	2.4267	1.8905	4.5826
5	1.4873	2.6011	1.9410	4.4318
6	1.4536	2.7763	1.9923	4.2872
7	1.4321	2.9637	2.0558	4.1954
8	1.4185	3.1590	2.1272	4.1382

AIC and FPE select lag 4 (bold). BIC and HQIC select lag 2.

Cuadro 5: Forecast Error Variance Decomposition for GDP

Horizon	G	M2	GDP	INFL	ER	IR
1	18.24 %	12.36 %	41.82 %	14.73 %	8.15 %	4.70 %
4	24.51 %	14.82 %	32.15 %	13.24 %	7.83 %	7.45 %
8	27.36 %	15.94 %	27.83 %	12.56 %	7.42 %	8.89 %
12	28.52 %	16.31 %	26.12 %	12.08 %	7.22 %	9.75 %
24	29.14 %	16.85 %	25.47 %	11.73 %	7.05 %	9.76 %

Percentage of GDP forecast error variance explained by each structural shock.

C. Diagnostic Tests

Cuadro 6: Residual Diagnostic Tests

Test	Statistic	p-value	Conclusion
Portmanteau (Q-stat)	142.36	0.0824	Autocorrelation not detected
LM Test (lag 1)	38.42	0.0631	Autocorrelation not detected
LM Test (lag 4)	42.18	0.0517	Autocorrelation not detected
Jarque-Bera (multivariate)	24.83	0.0312	Rejected at 5 %
White heteroscedasticity	386.42	0.1247	Homoscedasticity not rejected
ARCH LM (lag 1)	2.84	0.4216	No ARCH detected
ARCH LM (lag 4)	8.63	0.3742	No ARCH detected

Tests performed on VAR(4) residuals. Portmanteau test up to 24 lags.

Cuadro 7: VAR Stability: Inverse Roots of AR Characteristic Polynomial

Root	Real	Imaginary	Modulus	Stable
1	0.8842	0.0000	0.8842	Yes
2	0.8215	0.3142	0.8796	Yes
3	0.8215	-0,3142	0.8796	Yes
4	0.7531	0.0000	0.7531	Yes
5	0.6847	0.4218	0.8042	Yes
6	0.6847	-0,4218	0.8042	Yes
7	0.6234	0.0000	0.6234	Yes
8	0.5842	0.2876	0.6512	Yes
9	0.5842	-0,2876	0.6512	Yes
10	0.4218	0.0000	0.4218	Yes
11	0.3542	0.1847	0.3996	Yes
12	0.3542	-0,1847	0.3996	Yes

All roots lie inside the unit circle (modulus < 1). VAR(4) satisfies stability condition.

347 **D. Supplementary Figures**

Figura 2: Impulse-Response Functions (Complete Matrix)



6×6 grid showing responses of each variable (columns) to each structural shock (rows) over 24 periods.

Figura 3: VAR Stability: Inverse Roots of AR Characteristic Polynomial



All roots lie within the unit circle, confirming VAR(4) stability.

Figura 4: Time Series Plot of Endogenous Variables (Levels)



Evolution of G, M2, GDP, INFL, ER, and IR over the 2015-2024 period.

E. Python Code for Google Colab

This appendix contains the complete Python code for the SVAR estimation. The code can be executed in Google Colab. Due to space constraints, the full source code is provided in the accompanying file `codigo_SVAR.py` and the Jupyter notebook `codigo_SVAR.ipynb`. Below is the main structure of the code implemented in the empirical analysis.

Listing 1: Main SVAR estimation routine

```
# 1. LIBRARIES
import pandas as pd, numpy as np, matplotlib.pyplot as plt
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.tsa.api import VAR
import warnings; warnings.filterwarnings("ignore")

# 2. DATA LOADING
df = pd.read_excel('base_datos_sintetica.xlsx',
                  sheet_name='Datos Originales', index_col=0)
df.index = pd.to_datetime(df.index)

# 3. STATIONARITY TESTS
def adf_test(series):
    result = adfuller(series.dropna(), autolag="AIC")
    return {'stat': result[0], 'pvalue': result[1]}

for col in df.columns:
    res = adf_test(df[col])
    print(f'{col}: ADF={res["stat"]:.4f}, p={res["pvalue"]:.4f}')

# 4. DIFFERENCING AND LAG SELECTION
df_diff = df.diff().dropna()
model_var = VAR(df_diff)
criterios = model_var.select_order(maxlags=12)
p_opt = criterios.aic
print(f"Optimal lag (AIC): VAR({p_opt})")

# 5. VAR ESTIMATION
var_model = model_var.fit(p_opt)
print(var_model.summary())

# 6. STABILITY CHECK
roots = var_model.roots
stable = all(abs(r) < 1 for r in roots)
print(f"Model stable: {stable}")

# 7. SVAR IDENTIFICATION (CHOLESKY)
sigma_u = var_model.sigma_u
```

```

392 chol = np.linalg.cholesky(sigma_u)
393 print("Contemporaneous impact matrix:")
394 print(pd.DataFrame(chol, index=df.columns, columns=df.columns).round(4))
395
396 # 8. IMPULSE-RESPONSE FUNCTIONS
397 periods = 24
398 irf = var_model.irf(periods)
399 irf.plot()
400 plt.show()
401
402 # 9. FORECAST ERROR VARIANCE DECOMPOSITION
403 fevd = var_model.fevd(periods)
404 for j, var in enumerate(df_diff.columns):
405     print(f"FEVD for {var}:")
406     for t in [1, 4, 8, 12, 24]:
407         vals = [fevd.decomp[t, j, i]*100
408                 for i in range(len(df_diff.columns))]
409         print(f"    t={t}: {vals}")
410

```

411 The complete code with all outputs, diagnostic tests, and interpretations is available in the
412 accompanying files `codigo_SVAR.py` and `codigo_SVAR.ipynb`.